

AI 110: Foundations of AI Engineering

For CodePath Direct-to-Student

About This Syllabus

This document describes CodePath's course design, content, and typical course policies.

Your specific cohort may have variations in schedule, pacing, or policies. Check your CodePath Course Portal and welcome email for cohort-specific details including course dates, assignment deadlines, and instructor contact information.

When cohort-specific information differs from what's described in this syllabus, your cohort's information applies.

Course Overview

Course title and description

AI 110: Foundations of AI Engineering

A foundational course where students strengthen computer science fundamentals while programming in AI-assisted workflows, learning to code, debug, and reason critically with AI as a collaborator.

Course goals/learning outcomes

By the end of this course, you will be able to:

- Build confidence using AI tools for coding, debugging, and exploring new technologies
- Apply CS fundamentals—data structures, algorithms, and OOP—directly in AI-assisted development environments
- Demonstrate critical oversight and technical fluency in AI-augmented programming
- Build and present an application with responsible AI integration
- Develop awareness of tools and processes used in open source collaboration

Prerequisites/expected prior knowledge

- Be pursuing a course of study related to computer science or software
- Have completed at least one foundational programming course (e.g., Intro to CS or equivalent) or demonstrated proficiency in a language such as Python or Java
- Have an understanding of: loops, conditionals, arrays, data types, functions

Software/platform requirements

You must have access to a system with the ability to download and run Visual Studio Code.

- RAM: 8 GB RAM minimum, 16 GB RAM recommended
- Hard Drive Free Space: At least 10 GB of available disk space

A free GitHub account is required to access the CodePath platform.

Course Content & Structure

Topics covered

This course covers the following topics:

- **AI-Assisted Programming:** Writing, debugging, and reasoning with AI support from GitHub Copilot and Claude Code
- **Core CS Concepts:** Data structures, algorithms, and object-oriented programming in Python
- **Critical Evaluation:** Analyzing, refining, and explaining AI-generated code
- **Machine Learning Literacy:** Understanding supervised, unsupervised, and generative models
- **Applied AI Techniques:** Exploratory use of RAG, agentic workflows, fine-tuning, and guardrails
- **Collaboration and Open Source:** GitHub workflows and contribution practices

Key activities overview

Each week, you will participate in:

- **Interactive lectures:** Instructor-led sessions covering CS fundamentals and AI-assisted development techniques
- **Hands-on labs:** Guided practice writing, debugging, and refining code with AI assistance

- **Individual projects:** Weekly projects where you build applications using AI tools while maintaining human oversight and accountability
- **Portfolio project:** Design, build, and present an application demonstrating responsible AI integration

Textbook/required materials

All course materials will be provided through the CodePath platform. No textbook purchase is required.

Course schedule

 **The schedule below represents CodePath's standard timing.*

Specific timing for your cohort may vary. See your Course Portal for details.

This course consists of 5 modules delivered over 10 weeks, with each module spanning 2 weeks. Each week includes 2 hours of synchronous class time covering lectures, hands-on practice, and project work. In addition, students are expected to complete approximately 2–4 hours of project work per week outside of class.

Module	Week	Topic	Assignment Due
1	1	Thinking Like an AI-Native Programmer	
	2	Debugging and Refactoring with AI Tools	
2	3	Designing Systems with AI as a Partner	Project 1
	4	Algorithmic Thinking in AI Workflows	
3	5	How Machines Learn	Project 2
	6	Data, Bias, and Evaluation	
4	7	RAG and Connected Intelligence	Project 3
	8	Human + AI Collaboration	

5	9	Agentic Workflows, Fine-Tuned Models, and Testing	
	10	Presentation and Portfolio Showcase	Project 4

Assessment & Grading

Grading breakdown

 **The grading breakdown below reflects CodePath's standard structure for core course content. Your cohort may have **different requirements that affect your CodePath completer status**. See your Course Portal for details.*

Your grade in this course is based on individual project submissions:

Assignment	Points	Bonus Points
Project 1: Game Glitch Investigator	18	+10
Project 2: PawPal+	20	+10
Project 3: Music Recommender Simulation	21	+8
Project 4: Applied AI System	21	+8
Total	80	(+36)

Assessment structure

Each project includes both **required** and **stretch** features:

- **Required features:** Completion of all required features for a given project will earn 100% of the points for that project. If all required features are not completed, a grade will be assigned proportional to the point values for any completed features.
- **Stretch features:** Completion of stretch features will earn additional bonus points that may allow your percentage score to exceed 100%.

Detailed grading rubrics for each project, including specific point values for individual features, are available to students via the CodePath Course Portal.

Assignment submission details

All projects are submitted through the CodePath course platform. For each project, you will:

1. Complete the required (and optional stretch) features in your code editor
2. Push your code to GitHub
3. Update your project's README file using the provided template to document:
 - a. Completed required and stretch features
 - b. Screenshots or demo outputs showing your system's functionality
 - c. Reflection on AI collaboration, including helpful and flawed suggestions
 - d. Any known limitations or additional notes
4. Submit the link to your GitHub repository through the course platform

For the final project, you will also:

1. Include a system architecture diagram showing major components and data flow
2. Provide clear setup and usage instructions
3. Present your completed application, demonstrating responsible AI integration

Detailed submission instructions are provided with each assignment.

Course Policies

 **The policies in this section represent CodePath's standard recommendations.*

Specific policies for your cohort may vary. See your Course Portal for details.

Attendance expectations*

Regular attendance is expected at all class sessions. Students are allowed up to 2 absences during the course without penalty. Attendance is monitored throughout each session, and missing significant portions of class may count as an absence. Assignments remain due on their scheduled dates regardless of attendance.

Communication expectations*

During class sessions, students are expected to be present visually, verbally, and digitally:

- Keep your camera on and be visible from the shoulders up
- Use a laptop or computer (mobile devices are not permitted for class participation)
- Maintain a stable internet connection
- Be in a quiet environment that allows for active participation and engagement

Outside of class, communication happens through:

- **Slack Help Channel:** For programmatic and technical questions related to coursework
- **Email Support:** For general support, administrative questions, or personal concerns
- **Course Portal:** Check regularly for assignment updates and course resources

Students should allow up to 48 hours for responses to questions during the work week. For urgent technical issues during class time, use the designated help channels.

Late work policy*

All assignments must be submitted by their posted deadlines. If you miss a deadline, an automatic 48-hour extension is available without requesting permission. All work must be submitted within 48 hours of the original deadline to remain in good standing in the course

CodePath course completion*

To receive CodePath completer status and benefits, you must:

- Submit ALL required CodePath assignments
- Achieve a combined score of at least 70% on CodePath graded assignments

Students meeting these requirements will:

- Receive a digital CodePath certificate of completion

- Gain access to CodePath completer networks and benefits

Finding Additional Information

The following information is specific to your cohort and is available in your **CodePath Course Portal**:

- Instructor and teaching fellow contact information
- Office hours and technical support availability
- Specific course dates and assignment deadlines
- Communication channels (Slack workspace links, etc.)
- Student support resources and services

Access to your course will open **by the first day of class**.

AI 110: Foundations of AI Engineering

For University Partnerships

About This Syllabus

This document describes CodePath’s course design, content, and typical course policies.

Your specific section may have variations in schedule, pacing, or policies based on your university’s requirements. Check your University Learning Management System for section-specific details including course schedule, grading policies, and instructor information.

When university policies differ from CodePath’s recommendations in this syllabus, university policies apply.

Course Overview

Course title and description

AI 110: Foundations of AI Engineering

A foundational course where students strengthen computer science fundamentals while programming in AI-assisted workflows, learning to code, debug, and reason critically with AI as a collaborator.

Course goals/learning outcomes

By the end of this course, you will be able to:

- Build confidence using AI tools for coding, debugging, and exploring new technologies
- Apply CS fundamentals—data structures, algorithms, and OOP—directly in AI-assisted development environments
- Demonstrate critical oversight and technical fluency in AI-augmented programming
- Build and present an application with responsible AI integration
- Develop awareness of tools and processes used in open source collaboration

Prerequisites/expected prior knowledge

- Be pursuing a course of study related to computer science or software
- Have completed at least one foundational programming course (e.g., Intro to CS or equivalent) or demonstrated proficiency in a language such as Python or Java
- Have an understanding of: loops, conditionals, arrays, data types, functions

Software/platform requirements

You must have access to a system with the ability to download and run Visual Studio Code.

- RAM: 8 GB RAM minimum, 16 GB RAM recommended
- Hard Drive Free Space: At least 10 GB of available disk space

A free GitHub account is required to access the CodePath platform.

Course Content & Structure

Topics covered

This course covers the following topics:

- **AI-Assisted Programming:** Writing, debugging, and reasoning with AI support from GitHub Copilot and Claude Code
- **Core CS Concepts:** Data structures, algorithms, and object-oriented programming in Python
- **Critical Evaluation:** Analyzing, refining, and explaining AI-generated code
- **Machine Learning Literacy:** Understanding supervised, unsupervised, and generative models
- **Applied AI Techniques:** Exploratory use of RAG, agentic workflows, fine-tuning, and guardrails
- **Collaboration and Open Source:** GitHub workflows and contribution practices

Key activities overview

Each module, you will participate in:

- **Interactive lectures:** Instructor-led sessions covering CS fundamentals and AI-assisted development techniques
- **Hands-on labs:** Guided practice writing, debugging, and refining code with AI assistance

- **Individual projects:** Regular projects where you build applications using AI tools while maintaining human oversight and accountability
- **Portfolio project:** Design, build, and present an application demonstrating responsible AI integration

Textbook/required materials

All course materials will be provided through the CodePath platform. No textbook purchase is required.

Course schedule

 **The schedule below represents CodePath's standard university-partnered timing.*

Specific timing for your cohort may vary. See your University LMS for details.

This course consists of 5 modules delivered over a full semester (typically 15–16 weeks). Each module consists of 4 hours of synchronous class time, structured as two sessions covering lectures, hands-on practice, and project work. In addition, students are expected to complete approximately 2–4 hours of project work per week outside of class.

Sample Semester Schedule:

Week	Module	Topics Covered	Assignment Due
1	–	Course welcome and orientation	
2	1	Thinking Like an AI-Native Programmer	
3		Debugging and Refactoring with AI Tools	
4	–	Module 1 review	
5	2	Designing Systems with AI as a Partner	Project 1
6		Algorithmic Thinking in AI Workflows	
7	–	Spring break	

8	3	How Machines Learn	Project 2
9		Data, Bias, and Evaluation	
10	4	RAG and Connected Intelligence	Project 3
11		Professional development session	
12		Human + AI Collaboration	
13	–	Portfolio project: working session	
14	5	Agentic Workflows, Fine-Tuned Models, and Testing	
15		Portfolio project: working session	Project 4
16		Presentation and Portfolio Showcase	

Assessment & Grading

Grading breakdown

 **The grading breakdown below reflects CodePath’s standard structure for core course content. Your university may include **additional requirements (e.g., attendance, quizzes)** that affect your final university letter grade and CodePath completer status. See your University LMS for details.*

Your grade in this course is based on individual project submissions:

Assignment	Points	Bonus Points
Project 1: Game Glitch Investigator	18	+10
Project 2: PawPal+	20	+10

Project 3: Music Recommender Simulation	21	+8
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Assessment structure

Each project includes both **required** and **stretch** features:

- **Required features:** Completion of all required features for a given project will earn 100% of the points for that project. If all required features are not completed, a grade will be assigned proportional to the point values for any completed features.
- **Stretch features:** Completion of stretch features will earn additional bonus points that may allow your percentage score to exceed 100%.

Detailed grading rubrics for each project, including specific point values for individual features, are available to students via the CodePath Course Portal.

Assignment submission details

All projects are submitted through the CodePath course platform. For each project, you will:

1. Complete the required (and optional stretch) features in your code editor
2. Push your code to GitHub
3. Update your project's README file using the provided template to document:
 - a. Completed required and stretch features
 - b. Screenshots or demo outputs showing your system's functionality
 - c. Reflection on AI collaboration, including helpful and flawed suggestions
 - d. Any known limitations or additional notes
4. Submit the link to your GitHub repository through the course platform

For the final project, you will also:

1. Include a system architecture diagram showing major components and data flow
2. Provide clear setup and usage instructions
3. Present your completed application, demonstrating responsible AI integration

Detailed submission instructions are provided with each assignment.

Standard Course Policies

 *Course policies vary significantly by university and instructor. The descriptions below outline common expectations, but **your university's specific policies may differ**. Check your University LMS and course syllabus for details.*

Attendance expectations*

Class presence and participation are expected at all sessions. In most university partnerships:

- Attendance is tracked through various methods including quizzes, group activities, sign-in sheets, or participation points
- Students are expected to arrive on time, stay for the entire session, and actively contribute to in-class activities
- Students who repeatedly arrive late may have their participation grade lowered
- While not always required, it's good practice to notify your instructional team in advance if you must miss class

If you must miss class:

- For emergencies, notify your instructional team as soon as possible and provide relevant documentation
- For health-related absences, contact your university's Dean of Students Office for excused absence procedures
- You are responsible for obtaining class notes and announcements from classmates and catching up on missed materials
- All assignments remain due on their original deadlines unless you have received an approved extension

Check your university LMS for how attendance impacts your grade and to view your attendance scores.

Communication expectations*

During class sessions, students are expected to actively participate and engage with course content. This includes contributing to discussions, collaborating in group activities, and completing in-class work.

Outside of class, communication channels typically include:

- **CodePath Course Portal** for CodePath-specific resources, assignment submissions, and materials
- **Slack or other platforms** as specified by your instructional team for technical questions and peer collaboration
- **University LMS** (Canvas, Blackboard, iCollege, etc.) for course announcements, materials, and viewing grades

Response time expectations and preferred communication channels will be specified by your instructional team.

Late work policy*

For CodePath graded assignments:

All CodePath graded assignments are submitted through the CodePath course platform and graded by CodePath grading teams. Assignments must be submitted by their posted deadlines. If you miss a deadline, an automatic 48-hour extension is available without requesting permission. All CodePath assignments must be submitted within 48 hours of the original deadline.

For university-assigned coursework:

Your instructional team may assign additional coursework beyond CodePath assignments (such as quizzes, participation activities, or supplementary assignments). Late work policies for university-assigned work vary by cohort. Check your university LMS and course syllabus for policies on additional assignments.

CodePath course completion*

To receive CodePath completer status and benefits, you must:

- Submit ALL required CodePath assignments
- Achieve a combined score of at least 70% on CodePath graded assignments
- Complete **any additional requirements set by your university**

Students meeting these requirements will:

- Receive a digital CodePath certificate of completion
- Gain access to CodePath completer networks and benefits

Your university may have different requirements for earning course credit. Consult your university course syllabus for university-specific requirements.

Finding Additional Information

This syllabus describes CodePath's course design and content. Additional information specific to your university section is available in your University Learning Management System:

- Course section number (CRN) and credit hours
- Meeting times and classroom location
- Instructor name, contact information, and office hours
- Teaching assistant information
- University academic integrity policy
- Disability accommodations process and contact information
- Campus support services (tutoring, counseling, academic success centers)
- University-specific grading scale and requirements

Check your University LMS (Canvas, Blackboard, iCollege, etc.) for this information. When university policies differ from CodePath's recommendations in this syllabus, university policies apply.

AI 110 Syllabus Changelog – CodePath Direct-to-Student

Version	Date	Changes Made
v1.0	2026/02/19	Initial launch of new DTS-specific AI 110 syllabus
v1.0.1	2026/02/25	All students now receive instructions on installing both GitHub Copilot and Claude Code (PR #6537)
v1.0.2	2026/03/12	Instructor-facing guide for “Playlist Chaos” Tinker lab added to Week 1 instructor tab (PR #6557)
v1.0.3	2026/05/13	All GitHub Copilot-specific hints and instructions replaced with vendor-agnostic language, improving flexibility (PR #6649)
v1.0.4	2026/05/22	Replaced anthropomorphic AI language with technically accurate descriptions to help students build accurate mental models of AI (PR #6653)
v1.0.5	2026/05/27	- Added “(Lab)” and “(Project)” to Tinker and Show tab names to clarify their relationship to standard course formats (PR #6697) - Balanced stretch points allocation for Projects 1 and 3: P1 increased to +10, P3 reduced to +8 with two untaught stretch features removed (<i>Robust Test Suite, AI Model Comparison</i>).
v1.0.6	2026/06/18	Expanded the Module 3 sentiment-analysis lab (Tinker 3) with new ethics content: a new learning objective and activity steps distinguishing overt bias from covert compliance, plus two new model-card documentation sections.
v1.0.7	2026/07/06	Added a reference-solution exemplar repo to each of the five Tinker labs so students can compare against a complete working example
v1.1.0	2026/09/07	- Tinker labs restructured: new “Why this matters” research sections, “Key terms” lists, and pre-class environment setup, with each in-class part ending in an observable checkpoint. - Three new Tinker labs added: 1B (Split the Logic), 2B (Wire It Up), and 3B (Rank & Explain), each rehearsing the skills its paired project grades. “Key terms” vocabulary blocks and inline virtual-environment setup added to Projects 1 through 4.

AI 110 Syllabus Changelog – University Partnerships

Version	Date	Changes Made
v1.0	2026/02/19	Initial launch of new partner-specific AI 110 syllabus
v1.0.1	2026/02/25	All students now receive instructions on installing both GitHub Copilot and Claude Code (PR #6537)
v1.0.2	2026/03/12	Instructor-facing guide for “Playlist Chaos” Tinker lab added to Week 1 instructor tab (PR #6557)
v1.0.3	2026/05/13	All GitHub Copilot-specific hints and instructions replaced with vendor-agnostic language, improving flexibility (PR #6649)
v1.0.4	2026/05/22	Replaced anthropomorphic AI language with technically accurate descriptions to help students build accurate mental models of AI (PR #6653)
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